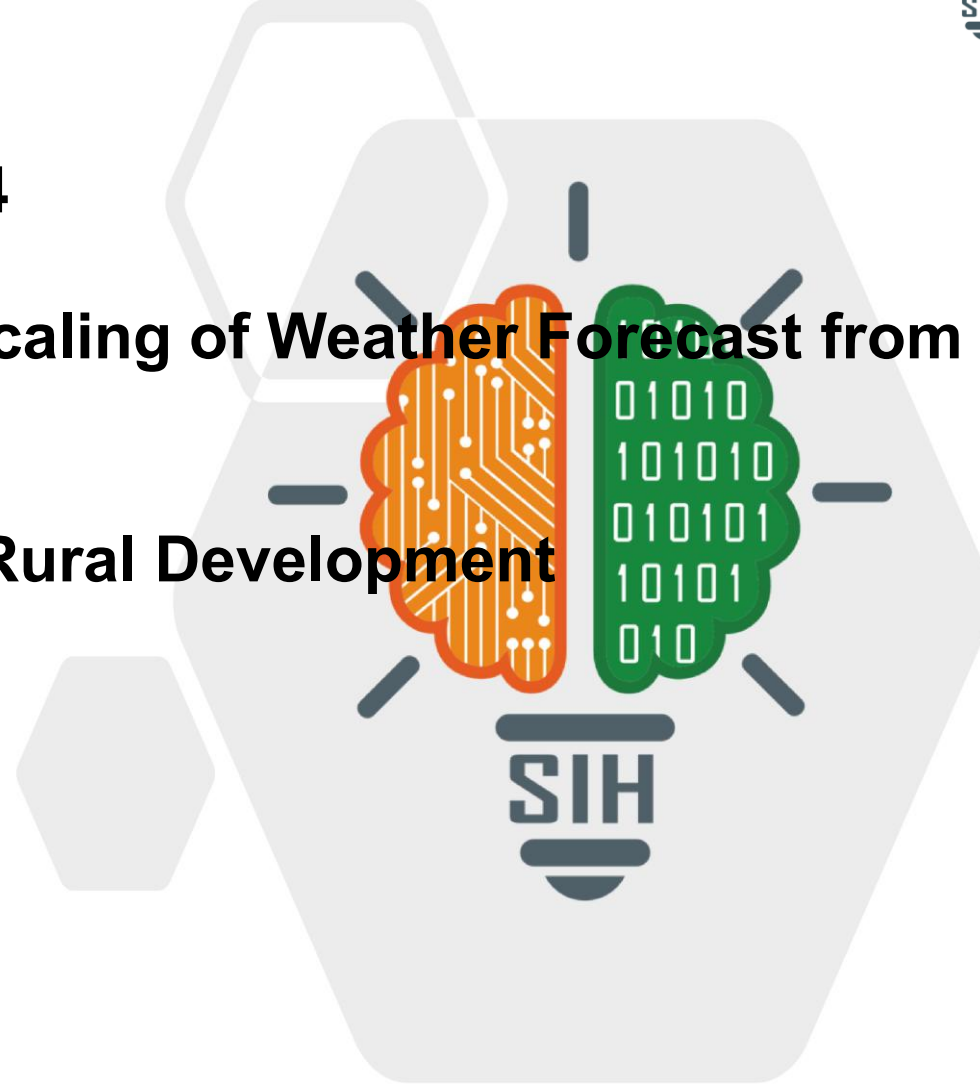


SMART INDIA HACKATHON 2026



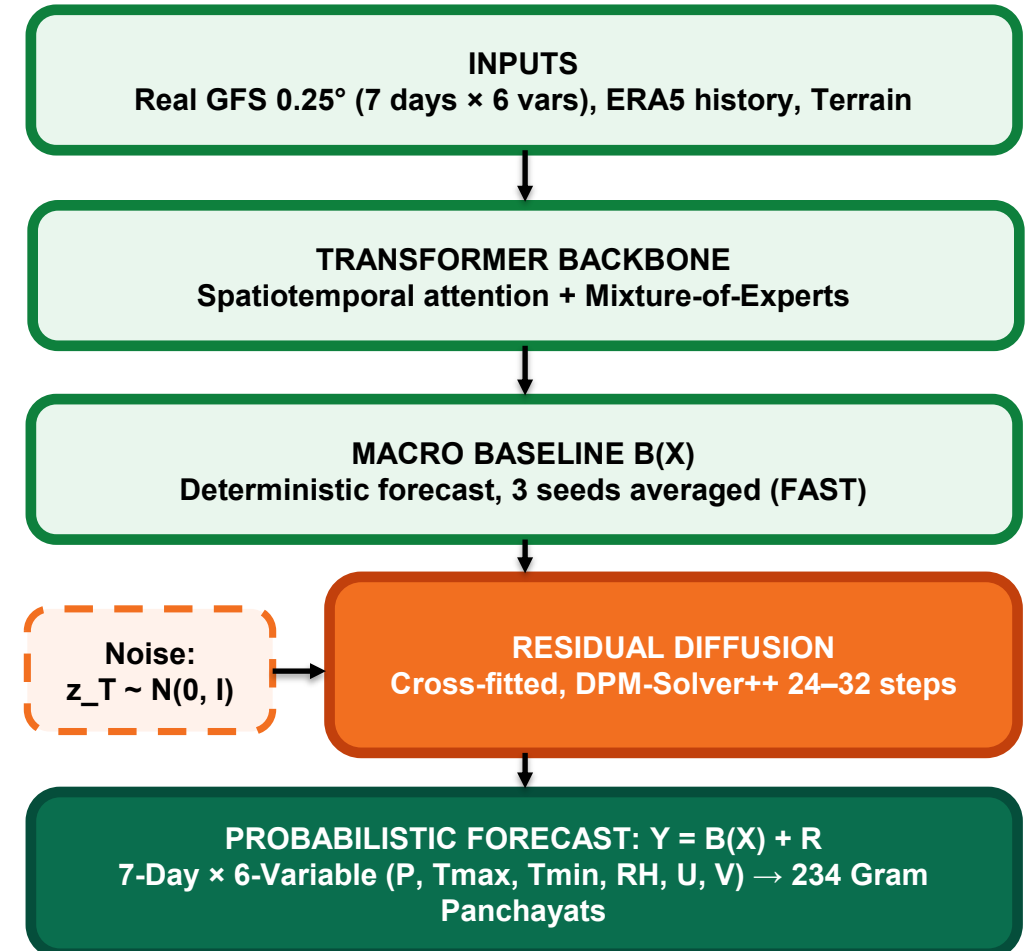
- **Problem Statement ID – SIH26074**
- **Problem Statement Title - Downscaling of Weather Forecast from Block level to Panchayat level**
- **Theme- Agriculture, FoodTech & Rural Development**
- **PS Category- Software**
- **Team ID - 173900**
- **Team Name - FAST&CURIOUS**



Problem Statement: A coarse 27 km (0.25°) GFS forecast gives one value for dozens of Gram Panchayats, missing local showers and terrain effects.

Proposed Solution:

- **Spatiotemporal Transformer Downscaling:** attention across space, time and the 6 variables, with a Mixture-of-Experts layer, turns the coarse forecast into a 0.05° (~5.5 km) field.
- **Cross-Fitted Residual Diffusion:** generates an ensemble of fine-scale detail on top of the transformer; each training season comes from a model that never saw it (rain CRPS –19 %).
- **Topography Coupling:** Copernicus GLO-30 elevation layers and a land mask capture ridge and valley effects.
- **Calibrated Uncertainty:** spread calibration and a rain tail cap; 5–95 % ranges cover 88 % of outcomes on the unseen 2023 monsoon.
- **Real Forecasts, Real Skill:** trained on 1,096 genuine GFS forecasts; on 2023 it cuts wet-day rain error by 20 % and Tmax error by 24 % versus raw GFS.

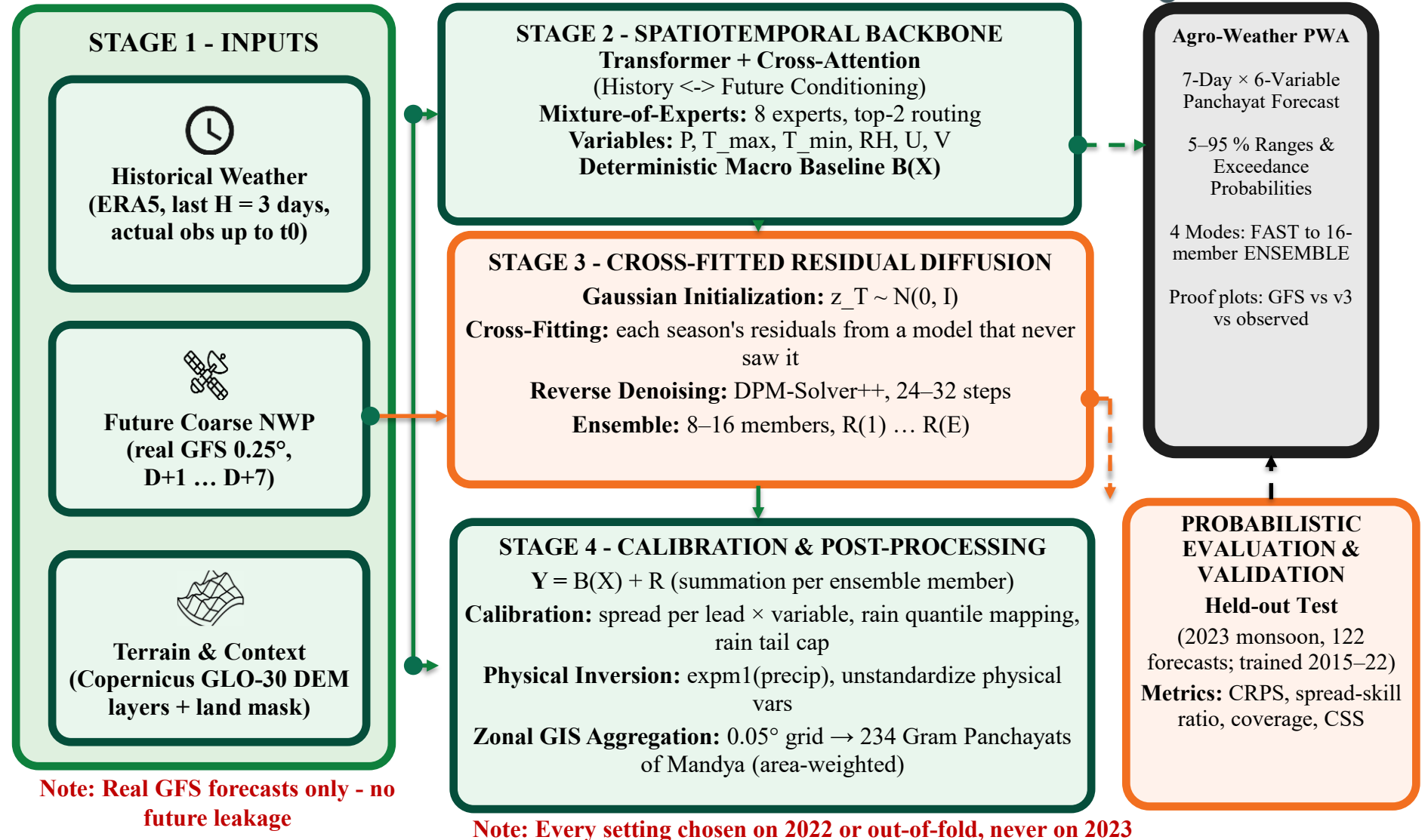


TECH STACK

- PyTorch
- Python, NumPy
- Xarray, Zarr, NetCDF
- ecCodes (GRIB)
- FastAPI, Uvicorn
- Pydantic
- Shapely
- Leaflet.js
- HTML5, CSS, JS
- Service Worker (PWA)
- SQLite
- Kaggle T4 GPUs

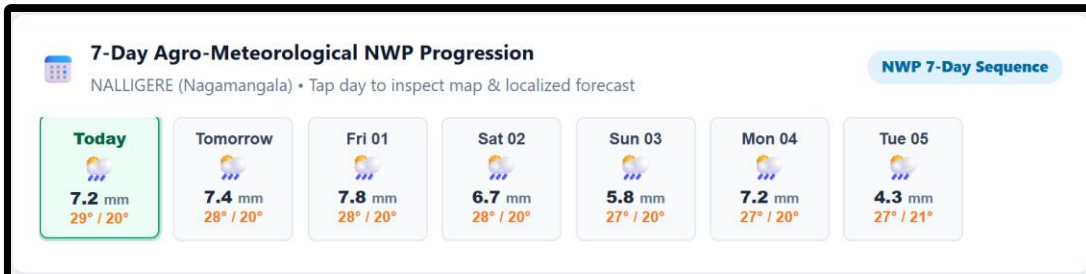
PERFORMANCE &
TESTING

- 0.17 s FAST forecast (T4)
- 8.3 s for 16 members
- < 1 GB GPU memory
- 234 GPs × 7 days × 6 vars



FEASIBILITY

- Runs in 0.17 s (FAST) to 8.3 s (16-member ensemble) on one T4 GPU, using < 1 GB of memory.
- Copernicus GLO-30 elevation layers and a land mask condition the model, capturing terrain effects at panchayat scale.
- Uses free, operational GFS forecasts as the input layer, avoiding the need to build a new numerical weather prediction system.



VIABILITY

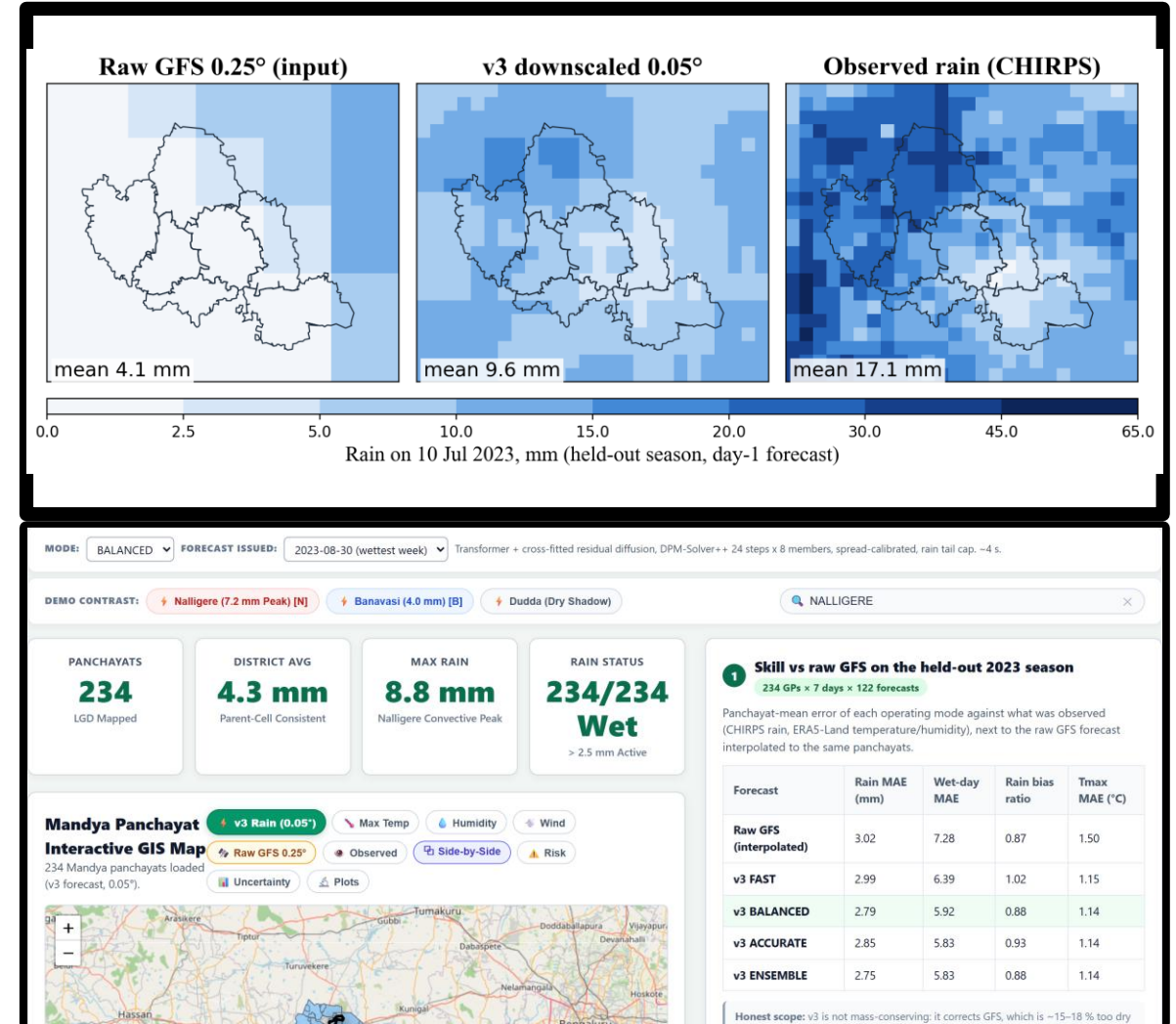
- Generates fine-resolution 7-day, 6-variable forecasts with calibrated uncertainty ranges.
- Converts GFS into Panchayat-level forecasts, maps and exceedance probabilities.
- Supports localized agro-weather advisories, Panchayat maps and spatial risk visualization.
- Input-agnostic: NCMRWF or IMD forecasts can replace GFS after retraining on their archives.
- Scales to any region with the same global data (GFS, CHIRPS, ERA5-Land, GLO-30) by retraining; the current model and demo cover all 234 Gram Panchayats of Mandya.

IMPACT

- Panchayat-level forecasts for all 234 Gram Panchayats of Mandya, validated on the unseen 2023 monsoon.
- Hyper-local weather information ($0.25^\circ \rightarrow 0.05^\circ$)
- Beats raw GFS at Panchayat level: wet-day rain error -20% , Tmax error -24% .

BENEFITS

- 6-Variable, 7-Day Forecasts: Joint precipitation, temperature, humidity and wind information provides richer agricultural context than rainfall alone.
- Calibrated Uncertainty: 5–95 % ranges cover 88 % of outcomes, and heavy-rain maps double the detection of ≥ 64.5 mm days.



DATASETS & PROJECT BASES

- NOAA GFS 0.25° operational forecasts (AWS Open Data, NCAR RDA ds084.1); model input.
- CHIRPS v2.0 0.05° daily precipitation (Funk et al., 2015); high-res rain target.
- Copernicus DEM GLO-30 30m DEM; terrain conditioning (elevation layers, land mask).
- ERA5 history & ERA5-Land 0.1° (Tmax, Tmin, RH, U, V) targets; Hersbach et al. 2020.
- Panchayat Boundaries & LGD Codes for Mandya, Karnataka (234 Gram Panchayats).

KEY METHODOLOGICAL REFERENCES

- Mardani et al. (2025) CorrDiff: residual corrective diffusion for km-scale downscaling. Commun. Earth Environ.
- Ho, Jain & Abbeel (2020) DDPM: basis for the residual diffusion component. NeurIPS 2020.
- Lu et al. (2022) DPM-Solver++: fast sampling of diffusion models. arXiv:2211.01095.
- Fedus, Zoph & Shazeer (2022) Switch Transformers: sparse Mixture-of-Experts. JMLR 23.
- Vandal et al. (2017) DeepSD: High-res climate projections via image super-resolution. KDD 2017.

Code, model weights & reports: github.com/rohzhedge26/SIH-26074-Fast-and-Curious